

19C ELECTROCHEMISTRY

UNIT 03: WHAT REACTIONS PROVIDE



CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 19

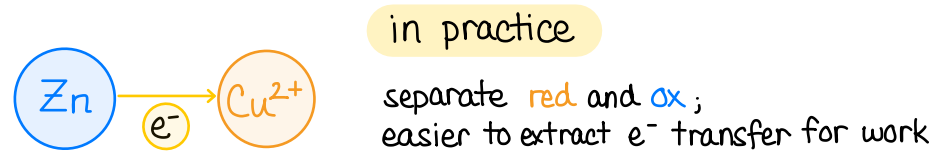
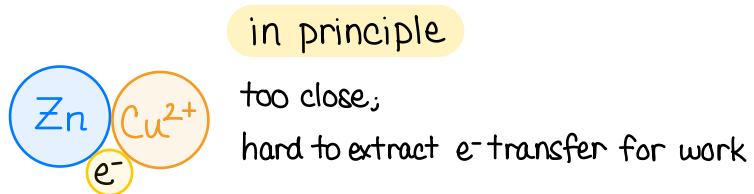
1. Define and assign oxidation states or oxidation numbers.
2. Define aspects of redox reactions: oxidation, reduction, oxidizing agents, and reducing agents.
3. Write half-reactions from overall redox reactions, and vice versa.
4. Balance redox reactions.
5. Describe the components of voltaic (or Galvanic) cells using appropriate terminology and notation.
6. Calculate cell potentials using tables of standard reduction potentials.
7. Compare the strengths of oxidizing and reducing agents using tables of standard reduction potentials.
8. Describe the relationship between the standard cell potential and the equilibrium constant.
9. Perform calculations involving standard cell potentials and equilibrium constants.
10. Calculate the effects of concentration on cell potential using the Nernst equation.
11. Describe how electricity can be used to drive redox reactions.
12. Perform stoichiometric calculations involving mass and current in electrolytic applications.

Voltaic Cells

19.4

Electrochemical cells

A system composed of two *separated* half-cells (corresponding to two half-reactions) that are connected by a salt bridge and a wire to allow for electron transfer.



There are two types:

1) Voltaic cells (also called *Galvanic cells*) $\Delta G_{\text{redox}} < 0$; $E_{\text{cell}}^{\circ} > 0$

- Use *spontaneous* redox reactions to create electricity (flow of electrons).

2) Electrolytic cells

$$\Delta G_{\text{redox}} > 0 ; E_{\text{cell}}^{\circ} < 0$$

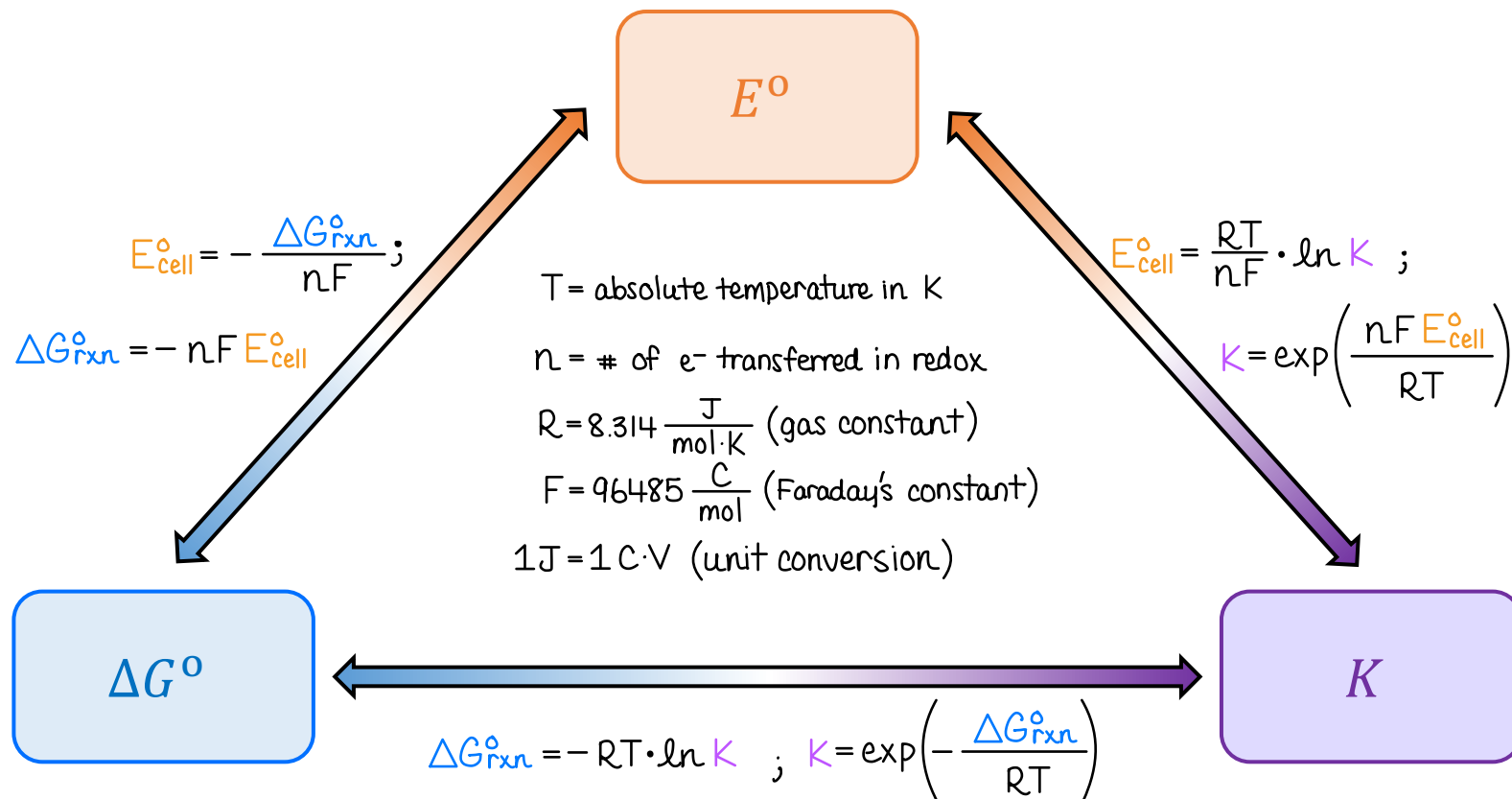
- Use electricity to cause a *nonspontaneous* redox reaction to occur.

Free Energy and Cell Potential

19.6

Free energies, equilibrium constants, and cell potentials

are all related: $\Delta G^\circ \leftrightarrow K \leftrightarrow E^\circ$



At standard ($^\circ$) conditions

1 M

1 atm

25 $^\circ\text{C}$

Spontaneous processes

$$\Delta G^\circ_{\text{rxn}} < 0$$

$$E^\circ_{\text{cell}} > 0$$

$$K > 1$$

Nonspontaneous processes

$$\Delta G^\circ_{\text{rxn}} > 0$$

$$E^\circ_{\text{cell}} < 0$$

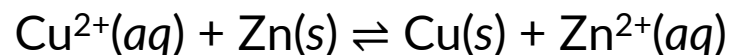
$$K < 1$$

EXAMPLE PROBLEM

Learning Objective(s): 19.9

notice spontaneous : $\Delta G_{\text{rxn}}^{\circ} < 0$, $E_{\text{cell}}^{\circ} > 0$, $K > 1$

The Daniell cell utilizes the Cu^{2+}/Cu and Zn^{2+}/Zn half-cells. Given the standard cell potential, calculate the values of $\Delta G_{\text{rxn}}^{\circ}$ (in kJ/mol) and K_c for the redox/cell reaction.



$$E_{\text{cell}}^{\circ} = +1.10 \text{ V}$$

$$\Delta G_{\text{rxn}}^{\circ} = -212 \text{ kJ/mol}$$

$$K_c = 1.55 \times 10^{37}$$

$$\Delta G_{\text{rxn}}^{\circ} = -nF E_{\text{cell}}^{\circ} \quad \Delta G_{\text{rxn}}^{\circ} = -RT \cdot \ln K \quad E_{\text{cell}}^{\circ} = \frac{RT}{nF} \cdot \ln K$$

$$\Delta G_{\text{rxn}}^{\circ} = -nF E_{\text{cell}}^{\circ}$$

$$= -(2) \left(96485 \frac{\text{C}}{\text{mol}} \right) (1.10 \text{ V})$$

$$= -2.122 \times 10^5 \frac{\text{C} \cdot \text{V}}{\text{mol}} \times \frac{1 \text{ J}}{1 \text{ C} \cdot \text{V}} \times \frac{1 \text{ kJ}}{1000 \text{ J}} \quad ; \quad 1 \text{ J} = 1 \text{ C} \cdot \text{V} \text{ (unit conversion)}$$

$$\Delta G_{\text{rxn}}^{\circ} = -212 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta G_{\text{rxn}}^{\circ} = -RT \cdot \ln K$$

$$K = \exp \left(-\frac{\Delta G_{\text{rxn}}^{\circ}}{RT} \right)$$

$$= \exp \left(-\frac{-2.122 \times 10^5 \frac{\text{J}}{\text{mol}}}{\left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (298.15 \text{ K})} \right)$$

$$K = 1.55 \times 10^{37}$$

The Nernst Equation and Concentration Cells

19.7

The effects of concentrations on cell potentials

- $\Delta G_{\text{cell}}^{\circ}$ and E_{cell}° are measured under standard conditions: $=1\text{ M}$ $=1\text{ atm}$
- ΔG_{cell} and E_{cell} are measured under **nonstandard conditions**: $\neq 1\text{ M}$ $\neq 1\text{ atm}$
- The Nernst equation relates E_{cell} , E_{cell}° , and the reaction quotient (Q).

substitute $\Delta G_{\text{rxn}} = -nFE_{\text{cell}}$

$$\Delta G_{\text{cell}} = \Delta G_{\text{cell}}^{\circ} + RT \cdot \ln Q$$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \cdot \ln Q$$

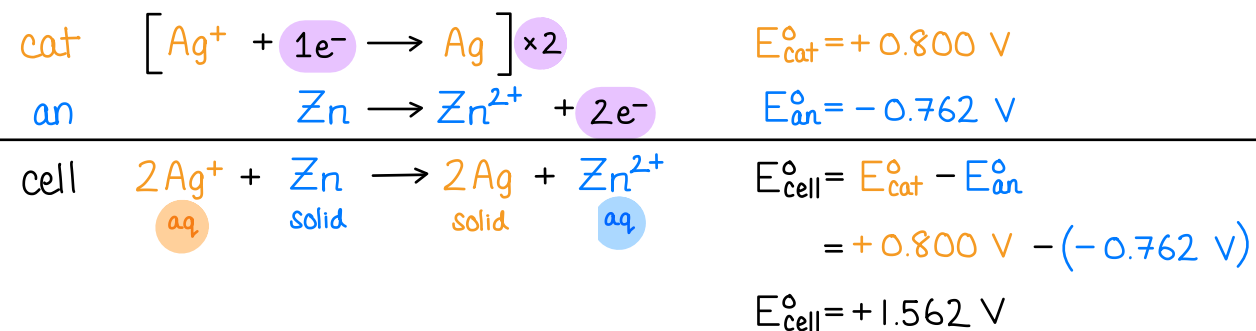
$R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$ (gas constant)
 T = absolute temperature in K
 Q = reaction quotient of redox equation
 $F = 96485 \frac{\text{C}}{\text{mol}}$ (Faraday's constant)
 n = # of e^{-} transferred in redox
 $1\text{ J} = 1\text{ C} \cdot \text{V}$ (unit conversion)

- **At equilibrium:** $\text{Rate}_{\text{for}} = \text{Rate}_{\text{rev}}$ $Q = K$ $\Delta G_{\text{rxn}} = 0$ $E_{\text{cell}} = 0$

IN-CLASS QUESTION 1

Learning Objective(s): 19.4, 19.6, 19.10

Calculate the potential (E_{cell} in V) of a zinc/silver cell in which the zinc electrode is immersed in $0.100 \text{ M Zn(NO}_3)_2(aq)$ and the silver electrode in $0.500 \text{ M AgNO}_3(aq)$ at 298 K.



Nernst equation: $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \cdot \ln Q$; $n=2$; $Q = \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$

$\neq 1 \text{ M}$

$$= +1.562 \text{ V} - \frac{\left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)(298 \text{ K})}{(2)\left(96485 \frac{\text{C}}{\text{mol}}\right)} \cdot \ln \left[\frac{0.100}{(0.500)^2} \right]$$

$$= +1.562 \text{ V} - (-0.01176 \text{ V})$$

$E_{\text{cell}} = +1.574 \text{ V}$

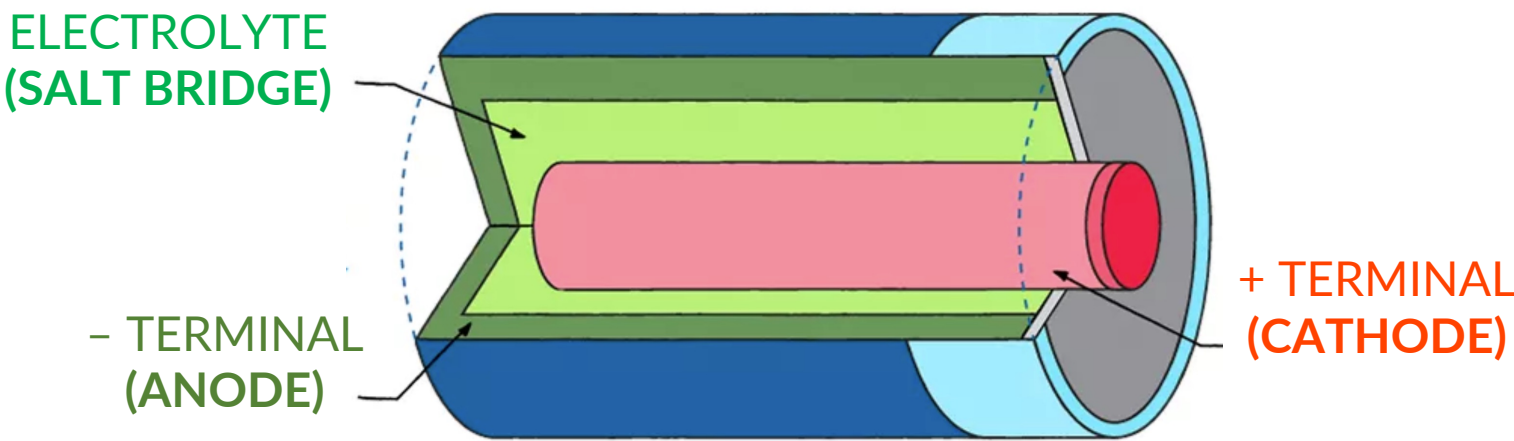
Reduction Half-Reaction								E_{red}° (V)
3	e^{-}	+	1	$\text{Au}^{3+} (aq)$	$\rightarrow$	1	$\text{Au} (s)$	+1.498 V
2	e^{-}	+	1	$\text{Cl}_2 (g)$	$\rightarrow$	2	$\text{Cl}^{-} (aq)$	+1.358 V
1	e^{-}	+	1	$\text{Ag}^{+} (aq)$	$\rightarrow$	1	$\text{Ag} (s)$	+0.800 V
1	e^{-}	+	1	$\text{Fe}^{3+} (aq)$	$\rightarrow$	1	$\text{Fe}^{2+} (aq)$	+0.771 V
1	e^{-}	+	1	$\text{Cu}^{+} (aq)$	$\rightarrow$	1	$\text{Cu} (s)$	+0.521 V
2	e^{-}	+	1	$\text{Cu}^{2+} (aq)$	$\rightarrow$	1	$\text{Cu} (s)$	+0.340 V
2	e^{-}	+	2	$\text{H}^{+} (aq)$	$\rightarrow$	1	$\text{H}_2 (g)$	0 V
2	e^{-}	+	1	$\text{Pb}^{2+} (aq)$	$\rightarrow$	1	$\text{Pb} (s)$	−0.126 V
2	e^{-}	+	1	$\text{Sn}^{2+} (aq)$	$\rightarrow$	1	$\text{Sn} (s)$	−0.138 V
2	e^{-}	+	1	$\text{Ni}^{2+} (aq)$	$\rightarrow$	1	$\text{Ni} (s)$	−0.257 V
2	e^{-}	+	1	$\text{Cd}^{2+} (aq)$	$\rightarrow$	1	$\text{Cd} (s)$	−0.403 V
2	e^{-}	+	1	$\text{Fe}^{2+} (aq)$	$\rightarrow$	1	$\text{Fe} (s)$	−0.447 V
3	e^{-}	+	1	$\text{Cr}^{3+} (aq)$	$\rightarrow$	1	$\text{Cr} (s)$	−0.744 V
2	e^{-}	+	1	$\text{Zn}^{2+} (aq)$	$\rightarrow$	1	$\text{Zn} (s)$	−0.762 V
3	e^{-}	+	1	$\text{Al}^{3+} (aq)$	$\rightarrow$	1	$\text{Al} (s)$	−1.662 V
2	e^{-}	+	1	$\text{Mg}^{2+} (aq)$	$\rightarrow$	1	$\text{Mg} (s)$	−2.372 V
1	e^{-}	+	1	$\text{Na}^{+} (aq)$	$\rightarrow$	1	$\text{Na} (s)$	−2.710 V

Battery Technology

Any battery can be considered a *portable voltaic/Galvanic cell*.

There many, many kinds of batteries.

There are really only three components to any battery, and you can optimize the anode, cathode, salt bridge, or all three.

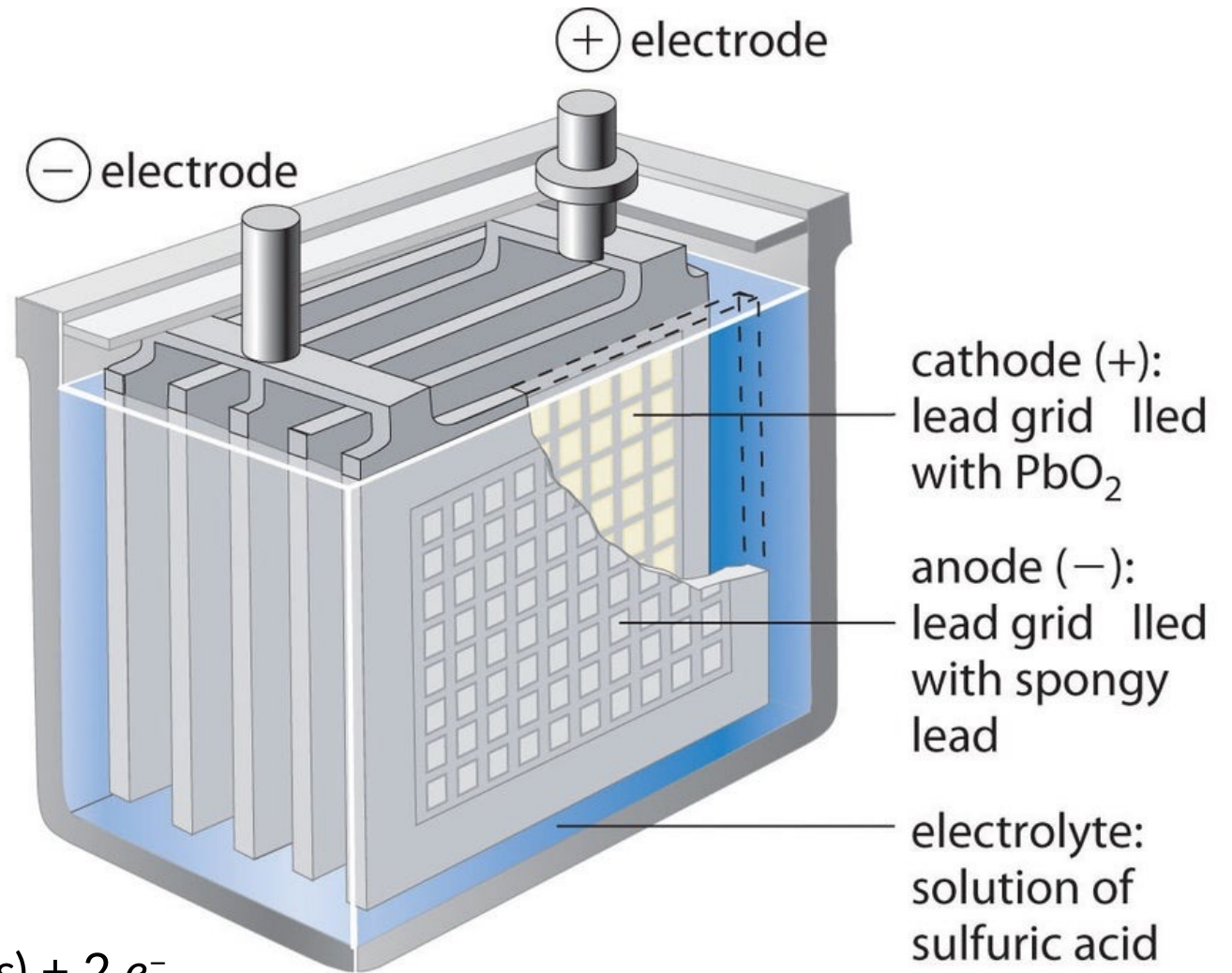


Primary batteries							
Chemistry	Anode (-)	Cathode (+)	Max. voltage, theoretical (V)	Nominal voltage, practical (V)	Specific energy (kJ/kg)	Elaboration	Shelf life at 25 °C, 80% capacity (months)
Zinc-carbon	Zn	C	1.6	1.2	130	Inexpensive.	18
Zinc-chloride	Zn	C	1.5			Also known as "heavy-duty", inexpensive.	
Alkaline (zinc-manganese dioxide)	Zn	MnO ₂	1.5	1.15	400-590	Moderate energy density. Good for high- and low-drain uses.	30
Nickel oxyhydroxide (zinc-manganese dioxide/nickel oxyhydroxide)			1.7			Moderate energy density. Good for high drain uses.	
Lithium (lithium-copper oxide) Li-CuO	Li	CuO	1.7			No longer manufactured. Replaced by silver oxide (IEC-type "SR") batteries.	
Lithium (lithium-iron disulfide) LiFeS ₂	Li	FeS ₂	1.8	1.5	1070	Expensive. Used in 'plus' or 'extra' batteries.	337 ^[47]
Lithium (lithium-manganese dioxide) LiMnO ₂	Li	MnO ₂	3.0		830-1010	Expensive. Used only in high-drain devices or for long shelf-life due to very low rate of self-discharge. 'Lithium' alone usually refers to this type of chemistry.	
Lithium (lithium-carbon fluoride) Li-(CF) _n	Li	(CF) _n	3.6	3.0			120
Lithium (lithium-chromium oxide) Li-CrO ₂	Li	CrO ₂	3.8	3.0			108
Lithium (lithium-silicon)	Li ₂₂ Si ₅						
Mercury oxide	Zn	HgO	1.34	1.2		High-drain and constant voltage. Banned in most countries because of health concerns.	36
Zinc-air	Zn	O ₂	1.6	1.1	1590 ^[48]	Used mostly in hearing aids.	
Zamboni pile	Zn	Ag or Au		0.8		Very long life. Very low (nanoamp, nA) current	>2,000
Silver oxide (silver-zinc)	Zn	Ag ₂ O	1.85	1.5	470	Very expensive. Used only commercially in 'button' cells.	30
Magnesium	Mg	MnO ₂	2.0	1.5			40

Secondary batteries					
Chemistry	Cell voltage	Specific energy (kJ/kg)	Energy density (kJ/liter)	Comments	
NiCd	1.2	140		Inexpensive. High-/low-drain, moderate energy density. Can withstand very high discharge rates with virtually no loss of capacity. Moderate rate of self-discharge. Environmental hazard due to Cadmium, use now virtually prohibited in Europe.	
Lead-acid	2.1	140		Moderately expensive. Moderate energy density. Moderate rate of self-discharge. Higher discharge rates result in considerable loss of capacity. Environmental hazard due to Lead. Common use: automobile batteries	
NiMH	1.2	360		Inexpensive. Performs better than alkaline batteries in higher drain devices. Traditional chemistry has high energy density, but also a high rate of self-discharge. Newer chemistry has low self-discharge rate, but also a ~25% lower energy density. Used in some cars.	
NiZn	1.6	360		Moderately inexpensive. High drain device suitable. Low self-discharge rate. Voltage closer to alkaline primary cells than other secondary cells. No toxic components. Newly introduced to the market (2009). Has not yet established a track record. Limited size availability.	
AgZn	1.86 1.5	460		Smaller volume than equivalent Li-ion. Extremely expensive due to silver. Very high energy density. Very high drain capable. For many years considered obsolete due to high silver prices. Cell suffers from oxidation if unused. Reactions are not fully understood. Terminal voltage very stable but suddenly drops to 1.5 volts at 70–80% charge (believed to be due to presence of both argentous and argentic oxide in positive plate; one is consumed first). Has been used in lieu of primary battery (moon buggy). Is being developed once again as a replacement for Li-ion.	
LiFePO ₄	3.3 3.0	360	790	Lithium-Iron-Phosphate chemistry.	
Lithium ion	3.6	460		Very expensive. Very high energy density. Not usually available in "common" battery sizes. Lithium polymer battery is common in laptop computers, digital cameras, camcorders, and cellphones. Very low rate of self-discharge. Terminal voltage varies from 4.2 to 3.0 volts during discharge. Volatile: Chance of explosion if short-circuited, allowed to overheat, or not manufactured with rigorous quality standards.	

Car Batteries

A car battery is made up of six lead-acid storage cells linked in series.



Anode (red): $\text{Pb(s)} + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{PbSO}_4(\text{s}) + 2 \text{e}^-$

Cathode (ox): $\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{PbSO}_4(\text{s}) + 2 \text{H}_2\text{O}(\ell)$

Overall cell: $\text{PbO}_2(\text{s}) + 2 \text{SO}_4^{2-}(\text{aq}) + 4 \text{H}^+(\text{aq}) + \text{Pb(s)} \rightarrow 2 \text{PbSO}_4(\text{s}) + 2 \text{H}_2\text{O}(\ell)$

Lithium-Ion Batteries



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John B. Goodenough

Prize share: 1/3



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M. Stanley Whittingham

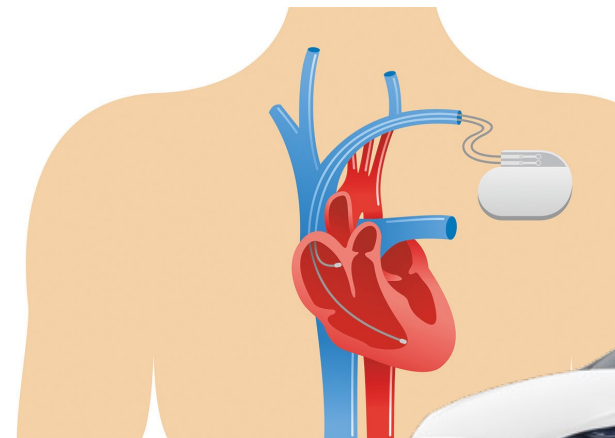
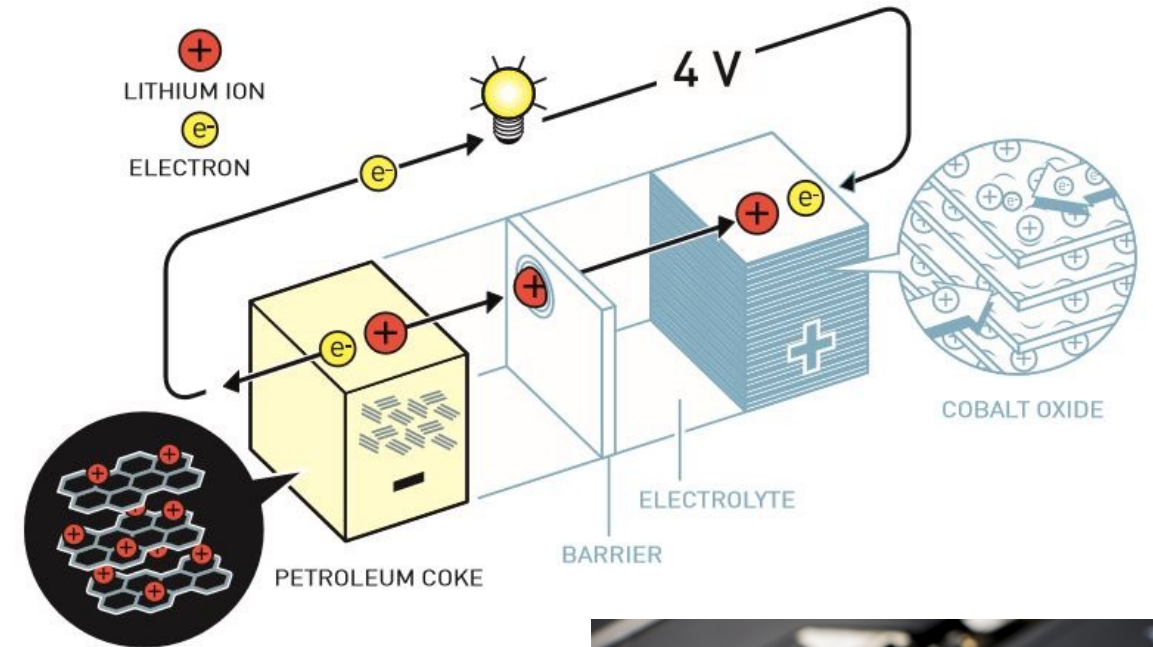
Prize share: 1/3



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Akira Yoshino

Prize share: 1/3



The Nobel Prize in Chemistry 2019 was awarded jointly to John B. Goodenough, M. Stanley Whittingham and Akira Yoshino "for the development of lithium-ion batteries"

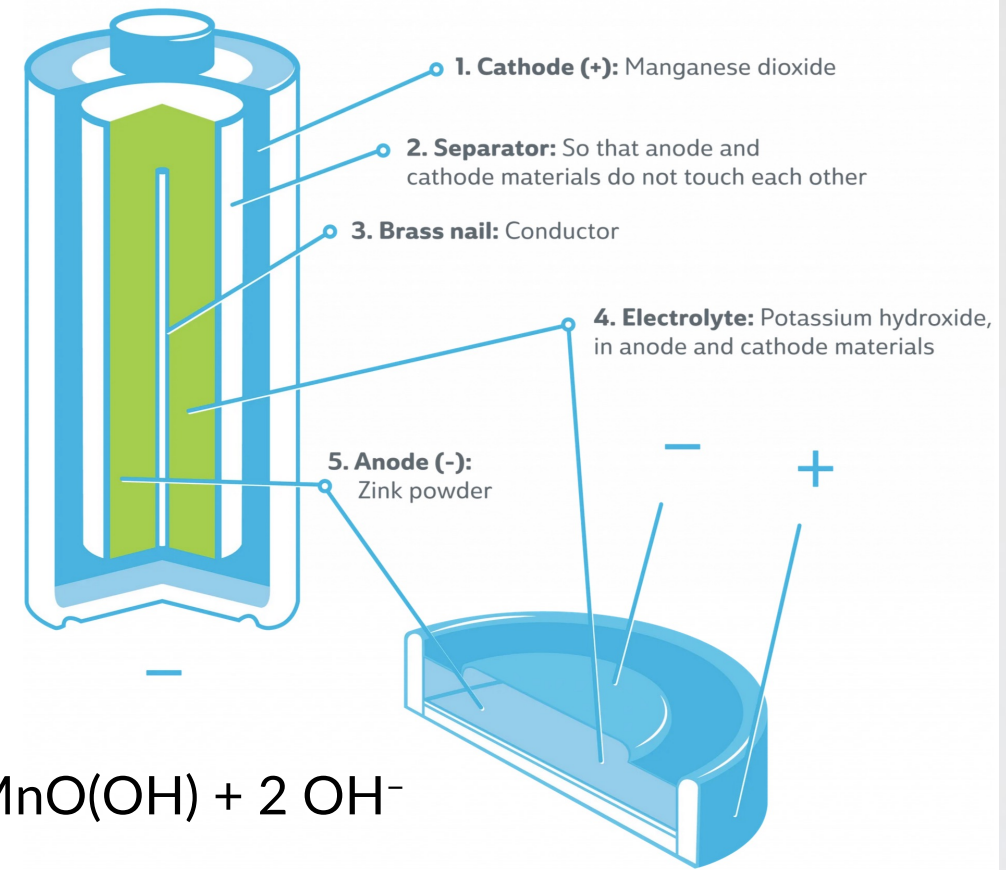
Alkaline Dry Cell Batteries

Common household batteries are called *alkaline dry cells* because the aqueous electrolytes are replaced by gels or pastes containing mobile ions.

Anode (ox): $\text{Zn} + 2 \text{OH}^- \rightarrow \text{Zn(OH)}_2 + 2 e^-$

Cathode (red): $\text{MnO}_2 + \text{H}_2\text{O} + e^- \rightarrow \text{MnO(OH)} + \text{OH}^-$

Overall cell: $\text{Zn} + 2 \text{MnO}_2 + 2 \text{H}_2\text{O} \rightarrow \text{Zn(OH)}_2 + 2 \text{MnO(OH)} + 2 \text{OH}^-$



Electrolytic Cells and Applications of Electrolysis

19.9

Electrochemical cells

A system composed of two half-cells (corresponding to the two half-reactions) that are connected by a salt bridge and a wire to allow for electron transfer.

There are two types:

- 1) Voltaic cells (also called *Galvanic cells*) $E_{\text{cell}}^{\circ} > 0 \text{ V}$ (positive)
 - Use *spontaneous* redox reactions to create electricity (a flow of electrons).
- 2) Electrolytic cells $E_{\text{cell}}^{\circ} < 0 \text{ V}$ (negative)
 - Use electricity to cause a *nonspontaneous* redox reaction to occur.

Electrolytic Cells and Applications of Electrolysis

19.9

Electrolytic cells: $E_{\text{cell}}^{\circ} < 0 \text{ V}$

- Apply electricity (current) to cause a **nonspontaneous redox** reaction to occur.
- An example of this is applying current to recharge certain batteries.

The most common application of electrolytic cells is called **electrolysis** or **electroplating**.

- Current is applied to an aqueous solution to **plate/coat metal onto another metal**.

↪ This happens at the **cathode** because it's where aq ions **reduced** to neutral metal solid!

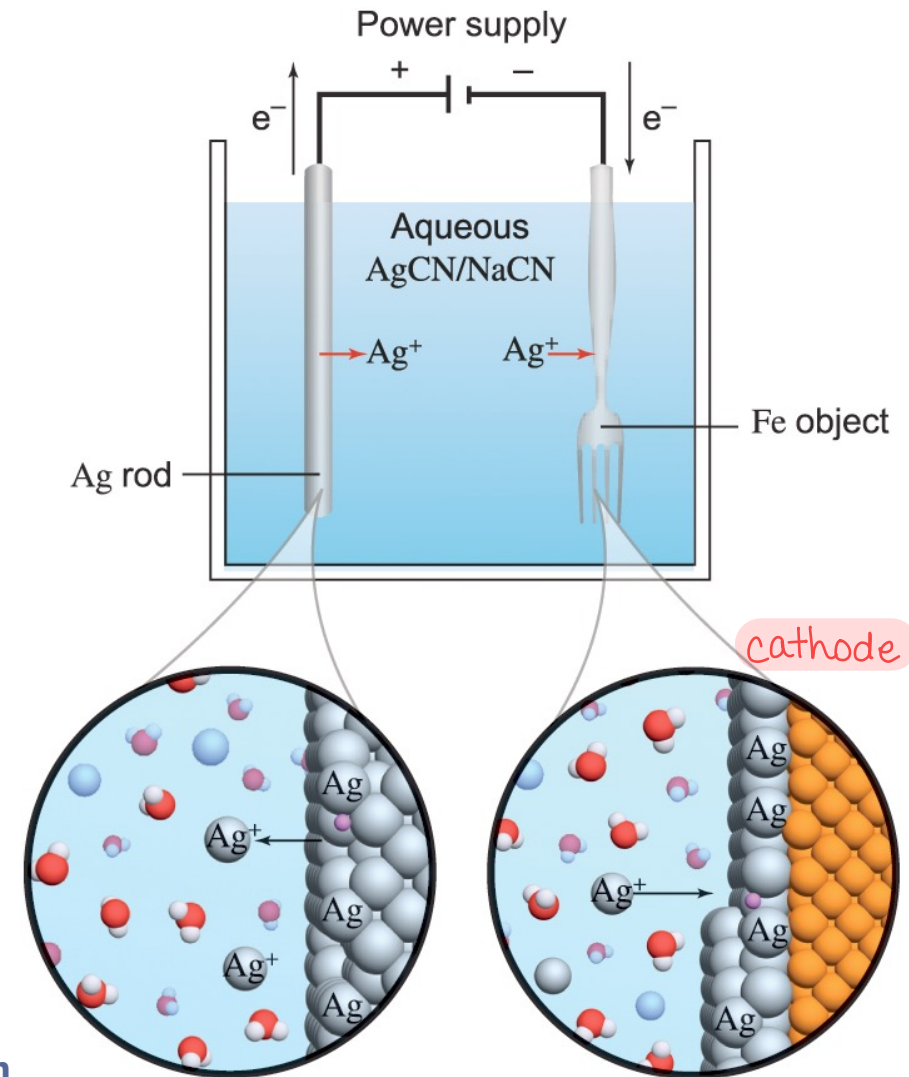


FIGURE 19.20 Electroplating silver from solution onto iron

Electrolytic Cells and Applications of Electrolysis

19.9

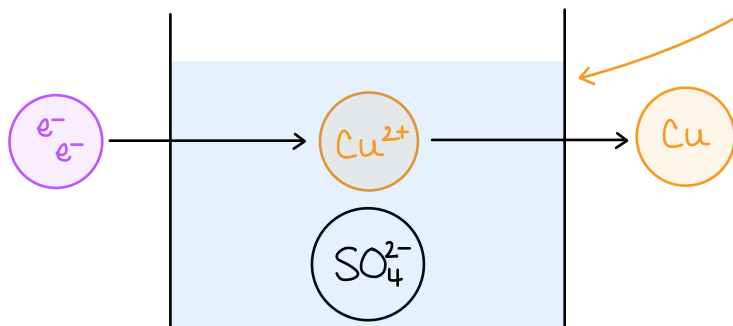
Calculations involving electrolysis

- The key is the total charge (q) transferred during electrolysis.
 - This is calculated by multiplying the current (I) and time (t).
$$q = I \cdot t$$
 - Current (I) is the charge (in Coulombs, C) per unit time.
 - Current (I) is measured in units of amperes (A) $\longleftarrow 1\text{ A} = \frac{1\text{ C}}{1\text{ s}}$
- Faraday's constant (F) is the charge of 1 mole of electrons (e^-). $1\text{ mol } e^- = 96485\text{ C}$
- As electrons are the carriers of charge in redox reactions, F allows the conversion between moles of e^- and the stoichiometry of the redox reaction of interest.
$$\text{or } F = \frac{96485\text{ C}}{1\text{ mol } e^-}$$

EXAMPLE PROBLEM

Learning Objective(s): 19.12

Calculate the mass (in grams) of copper metal that will be deposited by passing a **10.0 A** current through a solution of **copper(II) sulfate** for **3.00 hours**.



$$1 \text{ A} = \frac{1 \text{ C}}{1 \text{ s}}$$

TIP for electrolysis, only care about **cathode** because it's where aq ions are **reduced** to neutral metal solid!

$$\begin{array}{ccccccc}
 \text{time} & & \text{current} & & \text{Faraday's constant} & & \text{stoichiometry} & & \text{molar mass} & & \text{mass} \\
 3.00 \text{ hr} \times \frac{3600 \text{ s}}{1 \text{ hr}} \times \frac{10.00 \text{ C}}{1 \text{ s}} = 1.08 \times 10^5 \text{ C} & \times & \frac{1 \text{ mol e}^-}{96485 \text{ C}} = 1.119 \text{ mol e}^- & \times & \frac{1 \text{ mol Cu}}{2 \text{ mol e}^-} = 0.5597 \text{ mol Cu} & \times & \frac{63.546 \text{ g Cu}}{1 \text{ mol Cu}} = 35.6 \text{ g Cu}
 \end{array}$$

total charge passed through solution $\longrightarrow$ total number of mol of e^- transferred $\longrightarrow$ total number of mol of Cu from reduction

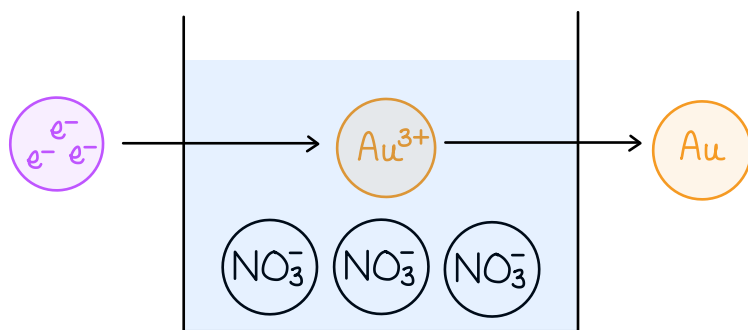
$$q = I \cdot t$$

$$F = \frac{96485 \text{ C}}{1 \text{ mol e}^-} \text{ (Faraday's constant)}$$

IN-CLASS QUESTION 2

Learning Objective(s): 19.12

Solid gold (Au) can be plated out of a solution containing $\text{Au}(\text{NO}_3)_3$. What mass of solid gold (in grams) can be plated after 10.0 minutes of electrolysis with a current of 5.50 amperes?



time

current

Faraday's
constant

stoichiometry

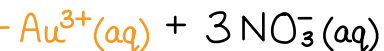
molar mass

mass

$$10.0 \text{ min} \times \frac{60 \text{ s}}{1 \text{ min}} \times \frac{5.50 \text{ C}}{1 \text{ s}} = 3.30 \times 10^3 \text{ C} \times \frac{1 \text{ mol } e^-}{96485 \text{ C}} = 0.0342 \text{ mol } e^- \times \frac{1 \text{ mol Au}}{3 \text{ mol } e^-} = 0.0114 \text{ mol Au} \times \frac{196.97 \text{ g Au}}{1 \text{ mol Au}} = 2.25 \text{ g Au}$$



TIP for electrolysis, only care about **cathode** because it's where aq ions are **reduced** to neutral metal solid!



$$1 \text{ A} = \frac{1 \text{ C}}{1 \text{ s}}$$

total charge passed through solution $\longrightarrow$ total number of mol of e^- transferred $\longrightarrow$ total number of mol of Au from reduction

$$q = I \cdot t$$

$$F = \frac{96485 \text{ C}}{1 \text{ mol } e^-} \text{ (Faraday's constant)}$$

BONUS QUESTION

Assessed on: *undetermined*

Calculate the values of E_{cell}° (in V), $\Delta G_{\text{rxn}}^{\circ}$ (in kJ/mol) and K_c for the voltaic/Galvanic cell composed of the Cu^{2+}/Cu and Ag^{+}/Ag half-cells.

Reduction Half-Reaction								E_{red}° (V)
3	e^{-}	+	1	$\text{Au}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Au}(\text{s})$	+1.498 V
2	e^{-}	+	1	$\text{Cl}_2(\text{g})$	$\rightarrow$	2	$\text{Cl}^{-}(\text{aq})$	+1.358 V
1	e^{-}	+	1	$\text{Ag}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Ag}(\text{s})$	+0.800 V
1	e^{-}	+	1	$\text{Fe}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Fe}^{2+}(\text{aq})$	+0.771 V
1	e^{-}	+	1	$\text{Cu}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Cu}(\text{s})$	+0.521 V
2	e^{-}	+	1	$\text{Cu}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Cu}(\text{s})$	+0.340 V
2	e^{-}	+	2	$\text{H}^{+}(\text{aq})$	$\rightarrow$	1	$\text{H}_2(\text{g})$	0 V
2	e^{-}	+	1	$\text{Pb}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Pb}(\text{s})$	−0.126 V
2	e^{-}	+	1	$\text{Sn}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Sn}(\text{s})$	−0.138 V
2	e^{-}	+	1	$\text{Ni}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Ni}(\text{s})$	−0.257 V
2	e^{-}	+	1	$\text{Cd}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Cd}(\text{s})$	−0.403 V
2	e^{-}	+	1	$\text{Fe}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Fe}(\text{s})$	−0.447 V
3	e^{-}	+	1	$\text{Cr}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Cr}(\text{s})$	−0.744 V
2	e^{-}	+	1	$\text{Zn}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Zn}(\text{s})$	−0.762 V
3	e^{-}	+	1	$\text{Al}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Al}(\text{s})$	−1.662 V
2	e^{-}	+	1	$\text{Mg}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Mg}(\text{s})$	−2.372 V
1	e^{-}	+	1	$\text{Na}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Na}(\text{s})$	−2.710 V